

# Zhenyu Liao

## Curriculum Vitae

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📄 Male, Chinese, born on 28/08/1992.

### Education

- |      |              |                                  |                                                        |
|------|--------------|----------------------------------|--------------------------------------------------------|
| 2019 | <b>Ph.D.</b> | Mathematics and Informatics      | L2S, CentraleSupélec, Paris-Saclay University, France. |
| 2016 | <b>M.Sc.</b> | Signal and Image Processing      | Paris-Saclay University, France.                       |
| 2014 | <b>B.E.</b>  | Optical & Electronic Information | Huazhong University of Science and Technology, China.  |

### Professional Experience

- 2021–present: **Associate Professor**, School of Electronic Information and Communications, Huazhong University of Science and Technology.
- 2020–2021: **Postdoctoral Scholar**, International Computer Science Institute (ICSI) and Department of Statistics, University of California, Berkeley, hosted by Michael Mahoney.

### Honors and Awards

- 2026: The **Springer Nature Editor of Distinction – Author Service Award 2026** for Statistics and Computing.
- 2025: IEEE Senior Member.
- 2025: Recipient of the **MATRIX-Simons Travel Grant**, MATRIX, Creswick, Australia.
- 2025: **CRM-Simons Visiting Scholar**, Centre de recherches mathématiques (CRM), Université de Montréal, for the thematic program **Mathematical Foundations of Data Science**. (*Unable to attend because of visa issues.*)
- 2024: **ANR-LabEx-CIMI Visiting Professor**, Centre International de Mathématiques et Informatique de Toulouse (CIMI), Toulouse, France.
- 2023: Recipient of a Talent Program of Hubei Province, China.
- 2021: Recipient of the Wuhan Youth Talent Program, Wuhan, China.
- 2021: Recipient of the East Lake Youth Talent Program Fellowship of HUST, Wuhan, China.
- 2019: Second Prize, **ED STIC Doctoral Prize**, Paris-Saclay University, France.

## Selected Publications

### Books

1. Romain Couillet and **Zhenyu Liao**. *Random Matrix Methods for Machine Learning*. Cambridge University Press, 2022. DOI: [10.1017/9781009128490](https://doi.org/10.1017/9781009128490).

### Papers in conference proceedings

1. Yanlei Liu and **Zhenyu Liao**. Dual-Attention Convolution Experts for Sparse Tensor Completion. In: *Machine Learning and Knowledge Discovery in Databases. Research Track (ECML PKDD)*. Springer Nature Switzerland, Sept. 2027, pp.450–467. [https://link.springer.com/chapter/10.1007/978-3-032-37670-1\\_26](https://link.springer.com/chapter/10.1007/978-3-032-37670-1_26).
2. Chiheb Yaakoubi, Cosme Louart, Malik Tiomoko, and **Zhenyu Liao**. Characterization of Gaussian Universality Breakdown in High-Dimensional Empirical Risk Minimization. In: *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. 2026. <https://arxiv.org/abs/2604.03146>.
3. Jiayu Bai, Danchen Yu, **Zhenyu Liao**, Tianqi Hou, Feng Zhou, Robert C. Qiu, and Zenan Ling. Diving into Kronecker Adapters: Component Design Matters. In: *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. 2026. <https://arxiv.org/abs/2602.01267>.
4. Yue Xu, Shuo Zhang, Gui Zhou, and **Zhenyu Liao**. Optimal Large-scale Finite-horizon Optimization via Random Matrix Theory. In: *2026 34th European Signal Processing Conference (EUSIPCO)*. 2026.
5. Yue Xu and **Zhenyu Liao**. An Improved Convergence Analysis of Gossip Methods for Large Random Graphs. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026, pp.22562–22566. DOI: [10.1109/ICASSP55912.2026.11461477](https://doi.org/10.1109/ICASSP55912.2026.11461477).
6. Yessin Moahker, Malik Tiomoko, Cosme Louart, and **Zhenyu Liao**. A Random Matrix Perspective of Echo State Networks: From Precise Bias-Variance Characterization to Optimal Regularization. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026, pp.20716–20720. DOI: [10.1109/ICASSP55912.2026.11464627](https://doi.org/10.1109/ICASSP55912.2026.11464627).

7. Jiuqi Wei, **Zhenyu Liao**, Ruoyu Han, Quanqing Xu, Chuanhui Yang, and Themis Palpanas. TaCo: Data-adaptive and Query-aware Subspace Collision for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (SIGMOD)* 4(3) (May 2026). doi: [10.1145/3802118](https://doi.org/10.1145/3802118).
8. Chengmei Niu, **Zhenyu Liao**, Zenan Ling, and Michael W. Mahoney. Fundamental Bias in Inverting Random Sampling Matrices with Application to Sub-sampled Newton. In: *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. Vol. 267. PMLR, July 2025, pp.46649–46692. <https://proceedings.mlr.press/v267/niu25c.html>.
9. Xiaoyi Mai and **Zhenyu Liao**. The Breakdown of Gaussian Universality in Classification of High-dimensional Mixtures. In: *International Conference on Learning Representations (ICLR)*. 2025. <https://openreview.net/forum?id=UrKbn51HjA>.
10. Jiuqi Wei, Xiaodong Lee, **Zhenyu Liao**, Themis Palpanas, and Botao Peng. Subspace Collision: An Efficient and Accurate Framework for High-dimensional Approximate Nearest Neighbor Search. *Proc. ACM Manag. Data (SIGMOD)* 3(1) (Feb. 2025). doi: [10.1145/3709729](https://doi.org/10.1145/3709729).
11. Wei Yang, Zhengyu Wang, Xiaoyi Mai, Zenan Ling, Robert Caiming Qiu, and **Zhenyu Liao**. Inconsistency of ESPRIT DoA Estimation for Large Arrays and a Correction via RMT. In: *2024 32nd European Signal Processing Conference (EUSIPCO)*. Aug. 2024, pp.2722–2726. doi: [10.23919/EUSIPCO63174.2024.10715081](https://doi.org/10.23919/EUSIPCO63174.2024.10715081).
12. Zenan Ling, Longbo Li, Zhanbo Feng, Yixuan Zhang, Feng Zhou, Robert C. Qiu, and **Zhenyu Liao**. Deep Equilibrium Models Are Almost Equivalent to Not-so-deep Explicit Models for High-dimensional Gaussian Mixtures. In: *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. 235. PMLR, July 2024, pp.30585–30609. <https://proceedings.mlr.press/v235/ling24a.html>.
13. Lingyu Gu, Yongqi Du, Yuan Zhang, Di Xie, Shiliang Pu, Robert Qiu, and **Zhenyu Liao**. "Lossless" Compression of Deep Neural Networks: A High-dimensional Neural Tangent Kernel Approach. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 35. Curran Associates, Inc., 2022, pp.3774–3787. [https://proceedings.neurips.cc/paper\\_files/paper/2022/hash/185087ea328b4f03ea8fd0c8aa96f747-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2022/hash/185087ea328b4f03ea8fd0c8aa96f747-Abstract-Conference.html).
14. Hafiz Tiomoko Ali, **Zhenyu Liao**, and Romain Couillet. Random matrices in service of ML footprint: ternary random features with no performance loss. In: *International Conference on Learning Representations (ICLR)*. 2022. <https://openreview.net/forum?id=qwULHx9zld>.
15. **Zhenyu Liao** and Michael W Mahoney. Hessian Eigenspectra of More Realistic Nonlinear Models. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 34. Curran Associates, Inc., 2021, pp.20104–20117. <https://papers.nips.cc/paper/2021/hash/a7d8ae4569120b5bec12e7b6e9648b86-Abstract.html>.
16. Michal Dereziński, **Zhenyu Liao**, Edgar Dobriban, and Michael Mahoney. Sparse sketches with small inversion bias. In: *Proceedings of Thirty Fourth Conference on Learning Theory (COLT)*. Vol. 134. PMLR, Aug. 2021, pp.1467–1510. <https://proceedings.mlr.press/v134/dereziński21a.html>.
17. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. Sparse Quantized Spectral Clustering. In: *International Conference on Learning Representations (ICLR)*. 2021. <https://openreview.net/forum?id=pBqLS-7KYAF>.
18. Fanghui Liu, **Zhenyu Liao**, and Johan Suykens. Kernel Regression in High Dimension: Refined Analysis beyond Double Descent. In: *Proceedings of The 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*. Vol. 130. PMLR, Apr. 2021, pp.649–657. <http://proceedings.mlr.press/v130/liu21b.html>.
19. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. A Random Matrix Analysis of Random Fourier Features: Beyond the Gaussian Kernel, A Precise Phase Transition, and the Corresponding Double Descent. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. Curran Associates, Inc., 2020, pp.13939–13950. <https://papers.nips.cc/paper/2020/hash/a03fa30821986dff10fc66647c84c9c3-Abstract.html>.
20. Michal Dereziński, Feynman T Liang, **Zhenyu Liao**, and Michael W. Mahoney. Precise expressions for random projections: Low-rank approximation and randomized Newton. In: *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 33. Curran Associates, Inc., 2020, pp.18272–18283. <https://papers.nips.cc/paper/2020/hash/d40d35b3063c11244fbf38e9b55074be-Abstract.html>.
21. Xiaoyi Mai, **Zhenyu Liao**, and Romain Couillet. A Large Scale Analysis of Logistic Regression: Asymptotic Performance and New Insights. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. May 2019, pp.3357–3361. doi: [10.1109/ICASSP.2019.8683376](https://doi.org/10.1109/ICASSP.2019.8683376).

22. Romain Couillet, **Zhenyu Liao**, and Xiaoyi Mai. Classification Asymptotics in the Random Matrix Regime. In: *The 26th European Signal Processing Conference (EUSIPCO)*. IEEE. Sept. 2018, pp.1875–1879. doi: [10.23919/EUSIPCO.2018.8553034](#).
23. **Zhenyu Liao** and Romain Couillet. The Dynamics of Learning: A Random Matrix Approach. In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80. PMLR, July 2018, pp.3072–3081. <http://proceedings.mlr.press/v80/liao18b.html>.
24. **Zhenyu Liao** and Romain Couillet. On the Spectrum of Random Features Maps of High Dimensional Data. In: *Proceedings of the 35th International Conference on Machine Learning (ICML)*. Vol. 80. PMLR, July 2018, pp.3063–3071. <http://proceedings.mlr.press/v80/liao18a.html>.
25. **Zhenyu Liao** and Romain Couillet. Random Matrices Meet Machine Learning: A Large Dimensional Analysis of LS-SVM. In: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. Mar. 2017, pp.2397–2401. doi: [10.1109/ICASSP.2017.7952586](#).

## Journal papers

1. **Zhenyu Liao** and Michael W. Mahoney. Random Matrix Theory for Deep Learning: Beyond Eigenvalues of Linear Models. *IEEE Signal Processing Magazine* **43**(2) (2026), 93–106. doi: [10.1109/MSP.2025.3618012](#).
2. Zhengyu Wang, Wei Yang, Xiaoyi Mai, Zenan Ling, **Zhenyu Liao**, and Robert C. Qiu. A Large-Dimensional Analysis of ESPRIT DoA Estimation: Inconsistency and a Correction via RMT. *IEEE Transactions on Signal Processing* **74** (2026), 2290–2303. doi: [10.1109/TSP.2026.3683673](#).
3. Junchao Lin, Zenan Ling, Zhanbo Feng, Jingwen Xu, Minxuan Liao, Feng Zhou, Tianqi Hou, **Zhenyu Liao**, and Robert C. Qiu. IGNN-Solver: A Graph Neural Solver for Implicit Graph Neural Networks. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2026), 1–18. doi: [10.1109/TPAMI.2026.3729092](#).
4. Zhanbo Feng, Yuanjie Wang, Jie Li, Fan Yang, Jiong Lou, Tiebin Mi, Robert. C. Qiu, and **Zhenyu Liao**. Robust and Communication-Efficient Federated Domain Adaptation via Random Features. *IEEE Transactions on Knowledge and Data Engineering* **37**(3) (2025), 1411–1424. doi: [10.1109/TKDE.2024.3510296](#).
5. Yacine Chitour, **Zhenyu Liao**, and Romain Couillet. A geometric approach of gradient descent algorithms in linear neural networks. *Mathematical Control and Related Fields* **13**(3) (2023), 918–945. doi: [10.3934/mcrf.2022021](#).
6. **Zhenyu Liao**, Romain Couillet, and Michael W. Mahoney. A random matrix analysis of random Fourier features: beyond the Gaussian kernel, a precise phase transition, and the corresponding double descent. *Journal of Statistical Mechanics: Theory and Experiment* **2021**(12) (Dec. 2021), 124006. doi: [10.1088/1742-5468/ac3a77](#).
7. **Zhenyu Liao** and Romain Couillet. A Large Dimensional Analysis of Least Squares Support Vector Machines. *IEEE Transactions on Signal Processing* **67**(4) (Feb. 2019), 1065–1074. doi: [10.1109/TSP.2018.2889954](#).
8. Cosme Louart, **Zhenyu Liao**, and Romain Couillet. A Random Matrix Approach to Neural Networks. *The Annals of Applied Probability* **28**(2) (Apr. 2018), 1190–1248. doi: [10.1214/17-AAP1328](#).

## Research Grants

1. 2026–2029: **PI**, National Natural Science Foundation of China General Program, *Compression of Large-scale Transformer-based Models with Random Matrix Methods: From Scaling Law to Dynamic Adaptive Model Compression* (NSFC-12571561), CNY 440K, with **Cosme Louart** as co-PI.
2. 2025–2030: **Co-PI**, National Key R&D Program of China (Youth Scientist Project), *Foundations of Machine Learning for Understanding Large Models* (2025YFA1018600), CNY 1M/3M. **PI**: **Jun Shu**, Xi'an Jiaotong University.
3. 2026–2028: **PI**, HUST Basic Research Support Program (Mathematics Track), *Theory and Methods for Large-scale Machine Learning via RMT* (2025BRAXB0004), CNY 380K.
4. 2024–2026: **PI**, Guangdong Key Lab of Mathematical Foundations for Artificial Intelligence Open Fund, *Generalization Theory for Transformer-based Models via Random Matrix Methods* (OFA00003), CNY 100K, with **Jeff Yao** as co-PI.
5. 2023–2025: **PI**, National Natural Science Foundation of China Youth Program, *Fundamental Limits of Pruning Deep Neural Network Models via Random Matrix Methods* (NSFC-62206101), CNY 300K.
6. 2022–2025: **Contributor**, National Natural Science Foundation of China grant, *Mathematical Foundations for Future Communications (Information Theory)* (NSFC-12141107), CNY 3M. **PI**: Robert C. Qiu.
7. 2021–2022: **PI**, CCF-Hikvision Open Fund, *Random Matrix Theory and Information Bottleneck for Neural Network Compression* (20210008), CNY 280K, with **Kai Wan** as co-PI.

8. 2018–2021: **Contributor**, NSF Research Grant, *Combining Stochastics and Numerics for Improved Scalable Matrix Computations* (NSF-1815054), USD 500K. PI: **Michael W. Mahoney**.
9. 2018–2021: **Contributor**, Programme d'Investissements d'avenir, *GSTATS IDEX DataScience Chair*, Université Grenoble Alpes, EUR 300K. PI: **Romain Couillet**.
10. 2014–2017: **Contributor**, French National Research Agency, *Random Matrix Theory for Large Dimensional Graphs* (ANR-14-CE28-0006), EUR 300K. PI: **Romain Couillet**.

## Professional Services

- Conference service for **ICML**, **NeurIPS**, **ICLR**, **AISTATS**, **IJCNN**, and **AAAI**.
- Associate Editor for Springer **Statistics and Computing**.
- Guest editor of the Special Topic "Random Matrix Theory and Its Applications" in **SCIENCE CHINA Mathematics**

## Organization of Scientific Activities

1. Co-organizer, with **Jianfeng Yao**, **Fan Yang**, **Jiang Hu**, **Dandan Jiang**, and **Shurong Zheng**, of the **2027 Annual Meeting on Random Matrix Theory and Applications**, January 14–17, 2027, Sanya, China.
2. Co-organizer, with **Zhigang Bao**, **Benoît Collins**, **Dong Yao**, **Jianfeng Yao**, and **Shurong Zheng**, the **TMRC-FRMTA Workshop**, August 16–22, 2026, Yunnan, China
3. **Sponsorship Co-chair** of the **2026 IEEE/CIC International Conference on Communications in China (ICCC)**, August 7–9, 2026.
4. Co-organizer, with **Jun Shu**, of the *Modern Machine Learning Theory and Methods* session at **CSML 2026**, August 7–9, 2026, Shanghai, China.
5. Co-organizer, with **Fanghui Liu** and **Lei Wu**, of the **TM10 session Theory Guides Practice for Large Models** at **CSIAM-BDAI 2026**, July 17–19, 2026, Xiamen, China.
6. Co-organizer, with **Rishabh Dudeja**, **Junjie Ma**, and **Arian Maleki**, of the **Universality and Dynamics in High-Dimensional Learning and Inference Workshop** at **ISIT 2026**, July 3, 2026, Guangzhou, China.
7. Co-organizer, with **Dirk Slock**, of the Special Session on *Large-Dimensional Structures and Methods for Signal Processing, Communication, and Machine Learning* at **ICASSP 2026**, May 2026, Barcelona, Spain.
8. Co-organizer, with **Pengkun Yang** and **Cong Fang**, of the **2025 Mathematical and Statistical Foundations of Foundation Models Workshop**, July 2025, Changbai Mountain, China.
9. Co-organizer of the **Founding Conference of the Random Matrix Theory and Applications Branch of CAAS**, January 2025, Changchun, China.
10. **General Co-chair** of the **10th EAI International Conference on IoT as a Service (EAI IoTaaS 2024)**, Wuhan, China.
11. Co-organizer of the **1st High-dimensional Learning Dynamics (HiLD) Workshop** at **ICML 2023**, with Mihai Nica, Courtney Paquette, Elliot Paquette, Andrew Saxe, and René Vidal.
12. Co-organizer of the **2022 HUST–Paris-Saclay Joint Workshop on Math for Data Science** with Yacine Chitour; more than 40,000 online attendees.

## Peer Reviewing

- **Research grants**: European Research Council (ERC), Natural Sciences and Engineering Research Council of Canada (NSERC), Israel Science Foundation (ISF), National Natural Science Foundation of China (NSFC).
- **Conferences**: **ICLR**, **ICML**, **NeurIPS**, **SODA**, **AISTATS**, **AAAI**, **ECAI**, **IEEE-ICASSP**, **IEEE-CAMSP**.
- **Journals**:
  - Mathematics, statistics, and data science: **SIAM Review (SIREV)**, **Foundations of Computational Mathematics (FoCM)**, **Physical Review X (PRX)**, **IEEE Transactions on Information Theory (TIT)**, **Journal of Mathematical Physics**, **Springer Statistics and Computing (STCO)**, **SIAM Journal on Scientific Computing (SISC)**, **Random Matrices: Theory and Applications (RMTA)**, **Latin American Journal of Probability and Mathematical Statistics (ALEA)**, **Annals of Applied Probability (AAP)**, **Electronic Communications in Probability (ECP)**, **Electronic Journal of Statistics (EJS)**, **Journal of Applied Mathematics and Computing (JAMC)**, **AIMS Mathematics**.
  - Machine learning and AI: **Journal of Machine Learning Research (JMLR)**, **IEEE Transactions on Pattern Analysis and Machine Intelligence (IEEE-TPAMI)**, **IEEE Signal Processing Letters (SPL)**, **IEEE Transactions on Signal Processing (IEEE-TSP)**, **IEEE Transactions on Multimedia (IEEE-TMM)**, **Engineering Applications of Artificial Intelligence (EAAI)**, **Frontiers of Computer Science**, **Neural Networks (NEUNET)**, **IEEE Transactions on Neural Networks and Learning Systems (IEEE-TNNLS)**, **IEEE Transactions on Industrial Informatics (IEEE-TII)**, **Transactions on Machine Learning Research (TMLR)**, **Pattern Recognition (PR)**, **Neural Processing Letters (NPL)**.